



University
of Glasgow

W1-04: Scaling (Horizontal vs. Vertical)

COMPSCI4064 (H) and COMPSCI5088 (M)

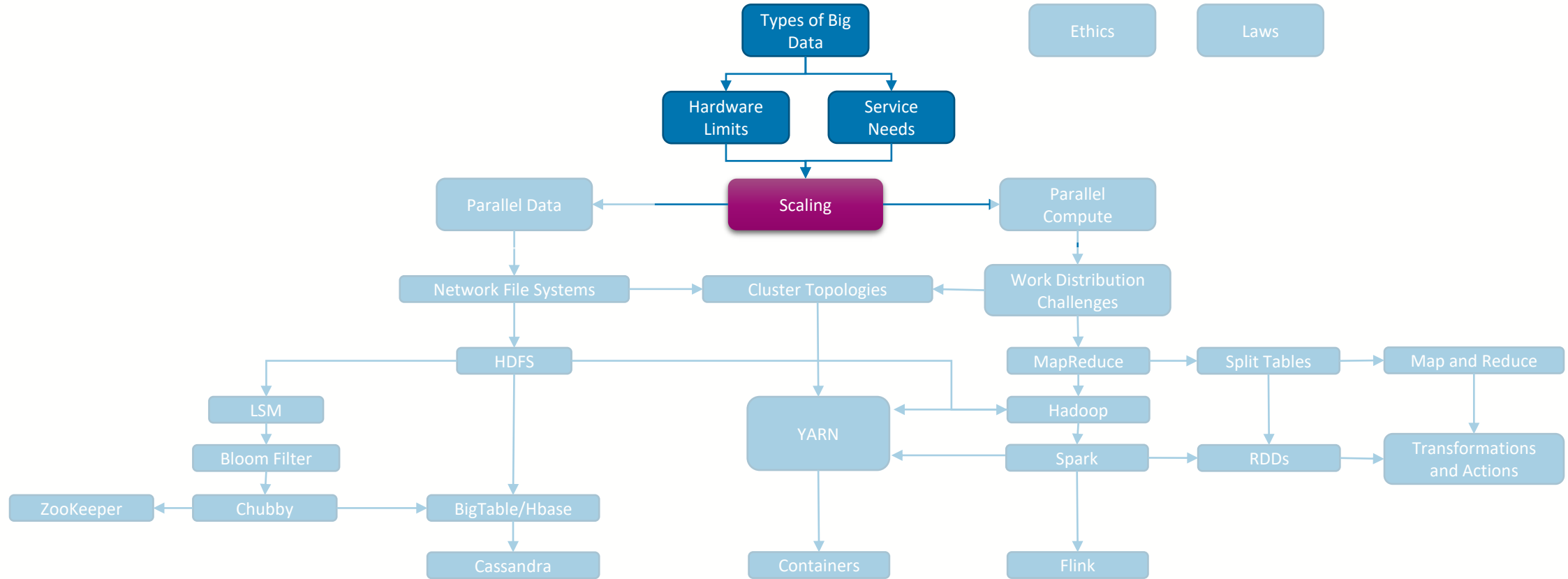
Dr. Richard McCreddie

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Where are we?





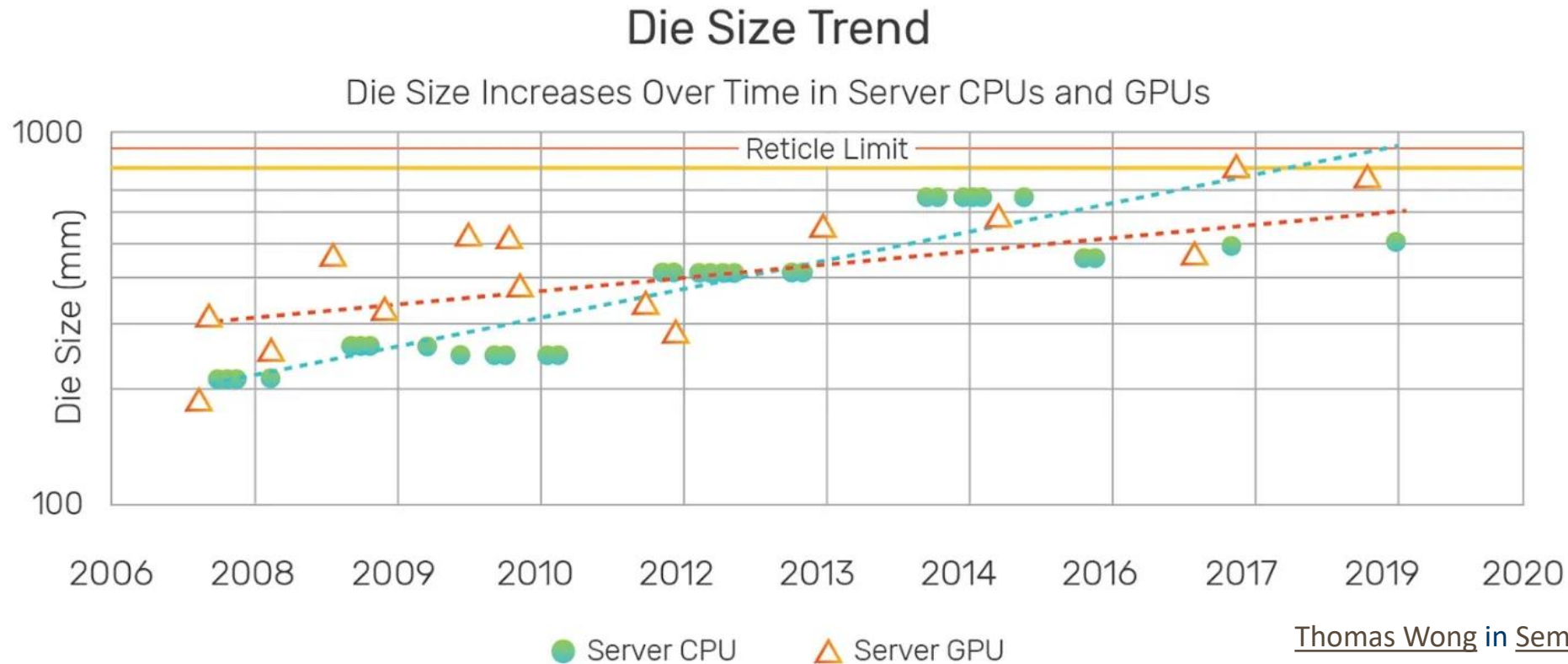
The Obvious Solution

- In simple terms, the solution to Big Data problems is always the same – **leverage more or faster hardware**
 - We refer to this as scaling-up our storage or processing
- In the 90's and early 2000's companies relied on what we now call **vertical scaling**
 - This is simply buying ever larger computers ('mainframes')
- However, this is **cost-ineffective**, as there are physical limits on what we can easily place on a single chip



Physical Limits – Compute

- We can only make CPUs and GPUs up to a set size (defined by the reticle limit)
 - We are already at that limit in some cases



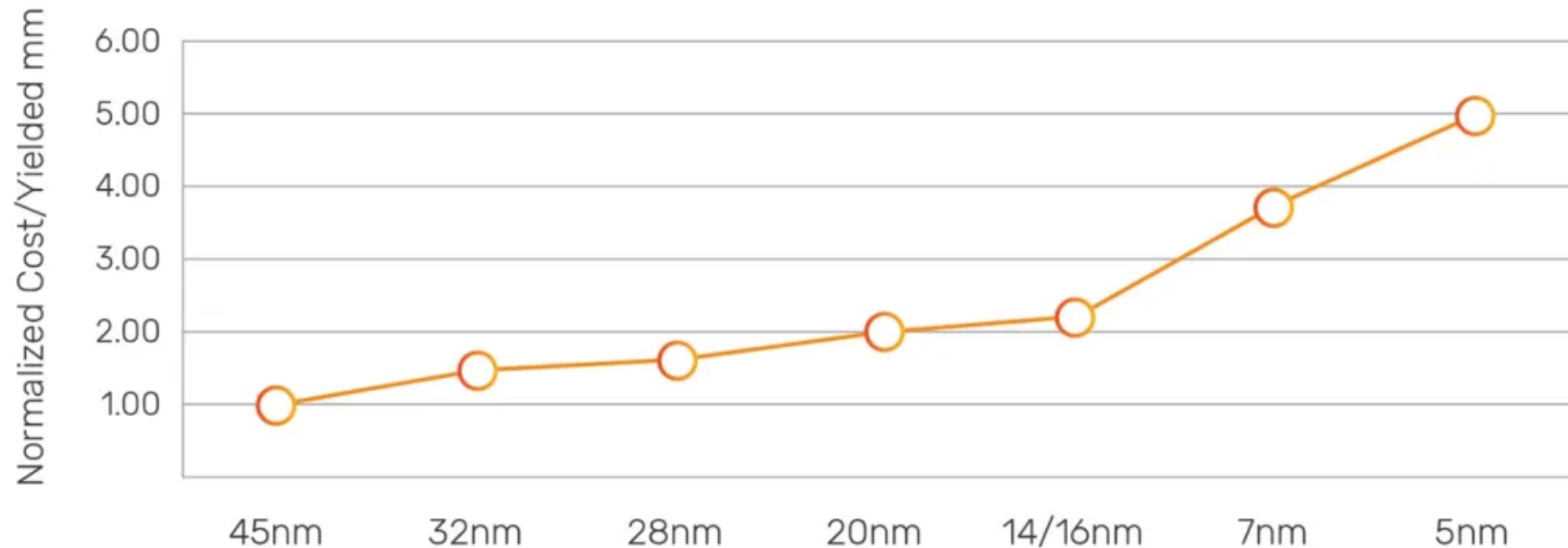


Physical Limits - Compute

- We can also make the transistors on a chip smaller, but this comes with additional cost
 - There are also questions on whether we have the technology to go much smaller

While Costs Continue to Increase

Cost per Yielded mm^2 for a 250mm^2 Die





Horizontal Scaling

- In the early 2000's there was significant research performed to distribute work over a cluster comprised of many low cost machines instead
 - This strategy is known as **horizontal scaling**
- Horizontal scaling is in theory infinitely scalable – you can simply add more machines
 - However what makes horizontal scaling tricky is that *leveraging more hardware requires the programmer or data scientist to program differently, and use specialised tools*
 - Also there are practical limits to horizontal scaling that need to be considered

The Big Data course is really a course about *Horizontal Scaling* of Compute and Storage, along with the tools that enable both



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